# Task 6 – Study of Website Hosting Errors and HTTP Status Codes

## Objective

1. Research and learn about the errors that arise regarding website hosting.
2. Study well about the errors within the error code range 300–500.
3. Note down the causes of these errors, as well as the troubleshooting methods.

## Research: Why Website Hosting Errors Occur

Website hosting errors are communicated through HTTP status codes — three-digit numbers a web server returns in response to a client's request. Understanding these codes helps identify whether a problem lies with redirection, the client's request, the server itself, or an upstream application. The first digit of the code indicates the general category:

| Range | Category | Meaning |
|---|---|---|
| 300–399 | Redirection | The requested resource requires another action or is available at another location |
| 400–499 | Client Errors | The request cannot be fulfilled due to a problem with the request or access |
| 500–599 | Server Errors | The server or an upstream service failed to process a valid request |

Note: 3xx responses are not strictly "errors" — they indicate redirection or caching behavior, though incorrect redirects can still cause real website problems.

## 2. Study of Error Codes (300–500 Range)

### 300 Range – Redirection

**301 – Moved Permanently**
The requested resource has permanently moved to another URL (e.g., old-sales.test → new-sales.test).

**302 – Found**
The requested resource is temporarily available at another URL.

**304 – Not Modified**
The requested resource hasn't changed since the client's cached version — the browser reuses its cached copy instead of re-downloading.

**307 – Temporary Redirect**
The resource is temporarily at another URL, while preserving the original HTTP method.

**308 – Permanent Redirect**
The resource has permanently moved, while preserving the original HTTP method.

**400 Range – Client-Side Errors**

**400 – Bad Request**
The server cannot understand or process the request because it is invalid.

**401 – Unauthorized**
Authentication is required, or the supplied credentials are invalid.

**403 – Forbidden**
The server understood the request but refuses to allow access.

**404 – Not Found (File Not Found Error)**
The server cannot find the requested resource.

**405 – Method Not Allowed**
The requested HTTP method (GET, POST, etc.) is not supported for that resource.

**408 – Request Timeout**
The server timed out while waiting for the client's request.

**409 – Conflict**
The request conflicts with the current state of the resource.

**429 – Too Many Requests**
The client sent too many requests within a specific period (rate limiting).

**500 Range – Server-Side Errors**

**500 – Internal Server Error**
A general server-side error occurred while processing the request.

**502 – Bad Gateway**
A reverse proxy/gateway received an invalid response from an upstream server.

**503 – Service Unavailable**
The server is temporarily unable to handle the request (overload, maintenance, or exhaustion).

**504 – Gateway Timeout**
A gateway/proxy did not receive a timely response from the upstream server.

**505 – HTTP Version Not Supported**
The server does not support the HTTP protocol version used in the request.

## 3. Causes and Troubleshooting Methods

### 301 / 308 – Moved / Permanent Redirect
*Causes:* domain changed, page URL changed, HTTP redirected to HTTPS.
*Troubleshooting:* check web server config, check .htaccess/Nginx config, verify destination URL, check for redirect loops.

### 302 / 307 – Found / Temporary Redirect
*Causes:* temporary maintenance, application-level redirect.
*Troubleshooting:* check redirect configuration, verify destination is reachable, remove redirect once no longer needed.

### 304 – Not Modified
*Causes:* browser caching, server caching, conditional requests.
*Troubleshooting:* hard refresh, clear browser cache, check server cache settings, verify latest file exists on server.

### 400 – Bad Request
*Causes:* incorrect URL syntax, malformed request, invalid parameters, corrupted cookies.
*Troubleshooting:* check the URL, clear cookies/cache, try incognito mode, check request parameters, inspect server logs.

### 401 – Unauthorized
*Causes:* missing/incorrect credentials, authentication config problems.
*Troubleshooting:* verify username/password, check Apache/Nginx auth config, verify account is active.

### 403 – Forbidden
*Causes:* incorrect file/directory permissions, SELinux restrictions, directory access rules, authorization rules.
*Troubleshooting:*

ls -l /var/www/html/
ls -Z /var/www/html/
getenforce
apachectl configtest
tail -f /var/log/httpd/error_log

(For Nginx: `tail -f /var/log/nginx/error.log`)

### 404 – Not Found (File Not Found Error)
*Causes:* incorrect URL, file doesn't exist, page deleted/moved, incorrect document root, incorrect VirtualHost config, missing application route.
*Troubleshooting:*

ls -l /var/www/html/
httpd -S

curl http://localhost/

If the page permanently moved, configure a 301 redirect.

## 405 – Method Not Allowed

*Causes:* incorrect HTTP method, application route doesn't support that method.
*Troubleshooting:* check application routes, verify allowed methods, check web server config, review application logs.

## 408 – Request Timeout

*Causes:* slow client connection, network problems, server timeout config.
*Troubleshooting:* check network connectivity, check timeout settings, review logs.

## 409 – Conflict

*Causes:* conflicting database/application operations, duplicate resource creation, version conflicts.
*Troubleshooting:* check application logs, check database records, verify resource state.

## 429 – Too Many Requests

*Causes:* rate limiting, excessive API calls, automated/abusive traffic.
*Troubleshooting:* check rate-limit config, identify traffic source, adjust limits if legitimate traffic is blocked.

## 500 – Internal Server Error

*Causes:* PHP/application errors, incorrect server config, incorrect .htaccess, missing dependencies, incorrect file permissions, resource limitations.
*Troubleshooting:*

sudo systemctl status httpd
sudo apachectl configtest
sudo tail -f /var/log/httpd/error_log

## 502 – Bad Gateway

*Causes:* backend application down, PHP-FPM unavailable, reverse proxy misconfiguration, unreachable upstream, incorrect upstream port.
*Troubleshooting:*

sudo systemctl status httpd
sudo systemctl status php-fpm
sudo ss -tulnp
sudo tail -f /var/log/httpd/error_log

## 503 – Service Unavailable

*Causes:* maintenance, server overload, stopped service, resource exhaustion, too many concurrent requests.
*Troubleshooting:*

sudo systemctl status httpd
sudo systemctl status php-fpm

top
free -h
df -h
sudo journalctl -xe

### 504 – Gateway Timeout

*Causes:* slow backend application, slow database query, overloaded backend, network issues, incorrect timeout config.
*Troubleshooting:* check backend status, check database performance, check CPU/memory, check network connectivity, review proxy timeout settings.

### 505 – HTTP Version Not Supported

*Causes:* client uses unsupported HTTP version, server software doesn't support requested protocol.
*Troubleshooting:* check client/server HTTP versions, update web server software if required.

## General Causes of Website Hosting Errors (Summary)

1. **Web Server Configuration** — incorrect VirtualHost, document root, redirect, or proxy configuration.
2. **File and Directory Permissions** — incorrect permissions causing 403 errors: `ls -l /var/www/html/`
3. **SELinux** — can block Apache from accessing files/network resources: `getenforce`, `ls -Z /var/www/html/`
4. **Firewall** — may block clients from reaching the server: `sudo firewall-cmd --list-services` (ports 80/443)
5. **DNS/Hostname Resolution** — incorrect DNS or /etc/hosts entries: `getent hosts example.com`
6. **Service Failure** — Apache, Nginx, PHP-FPM, or database service stopped: `systemctl status httpd/nginx/php-fpm/mariadb`
7. **Application Errors** — code errors or missing dependencies in PHP/Python/Node.js apps.
8. **Resource Exhaustion** — insufficient CPU/RAM/disk: `top`, `free -h`, `df -h`

## General Troubleshooting Method (Step-by-Step)

1. Check the URL — verify domain and path are correct.
2. Test connectivity — `ping <server-ip>`
3. Test HTTP response — `curl -I http://<server-ip>`
4. Check web server status — `systemctl status httpd` / `systemctl status nginx`
5. Check configuration — `apachectl configtest` / `nginx -t`
6. Check listening ports — `ss -tulnp`

7. Check firewall — `firewall-cmd --list-services`
8. Check permissions and SELinux — `ls -l`, `ls -Z`, `getenforce`
9. Check logs — Apache: `/var/log/httpd/error_log`, `access_log`; Nginx: `/var/log/nginx/error.log`, `access.log`
10. Check backend services — `systemctl status php-fpm`, `systemctl status mariadb`

**Quick Reference Table**

| Code | Name | Main Area |
|---|---|---|
| 301 | Moved Permanently | Permanent redirection |
| 302 | Found | Temporary redirection |
| 304 | Not Modified | Browser/server caching |
| 307 | Temporary Redirect | Temporary redirection |
| 308 | Permanent Redirect | Permanent redirection |
| 400 | Bad Request | Invalid client request |
| 401 | Unauthorized | Authentication required |
| 403 | Forbidden | Access denied |
| 404 | Not Found | Resource does not exist |
| 405 | Method Not Allowed | Unsupported HTTP method |
| 408 | Request Timeout | Client request timed out |
| 409 | Conflict | Resource/state conflict |
| 429 | Too Many Requests | Rate limiting |
| 500 | Internal Server Error | Server/application failure |
| 502 | Bad Gateway | Invalid upstream response |
| 503 | Service Unavailable | Service unavailable/overloaded |
| 504 | Gateway Timeout | Upstream timeout |
| 505 | HTTP Version Not Supported | Unsupported HTTP version |

## Conclusion

HTTP status codes provide an important method for diagnosing website hosting problems. The 300 range mainly deals with redirection and caching, the 400 range generally indicates problems with the client's request or access permissions, and the 500 range indicates failures on the server or upstream application side. For a Linux web server administrator, troubleshooting should include checking service status, configuration files, ports, firewall rules, DNS/hostname resolution, file permissions, SELinux contexts, system resources, and server/application logs. Understanding these status codes makes it easier to identify the location of a problem and apply the appropriate troubleshooting method.